

Ch2 $\Rightarrow$ Basic Notions of Q.M.

Ch2.1 $\rightarrow$ concept of Q.M.

Ch2.2 Hilbert space & operators (Infinite dim Linear Algebra).

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Some Example of Q.M. (harmonic oscillator)

How to Solve PDE in algebraic way

2.1. Generalities.

In Newtonian mechanics

$\Rightarrow$ position, momentum $\rightarrow$ exactly

In Schrödinger Eq / Q.M.

$\Rightarrow$ We can not measure position & momentum A B
simultaneously.

$\rightarrow$ Position, momentum, Energy, Angular momentum, ...
① Observable : Measurable Physical Quantities.

② State : Quantum particle

$\rightarrow$ Information position / momentum.

$\psi, \phi, \dots \Rightarrow$ States.

$\hat{p}, \hat{x}, \hat{H} \Rightarrow$ operators
 $\hat{p}$ momentum operator
 $\hat{x}$ position op-
 $\hat{H}$ Hamiltonian operator.

$\hat{p}\psi \Rightarrow$ momentum of the state

$$[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x}$$

$$= i\hbar$$

Momentum, position, $\Rightarrow$

$$\hat{x}\hat{p}\psi \neq \hat{p}\hat{x}\psi$$

Planck constant
 $\approx 10^{-34}$

Classical mech. $\Rightarrow$ $\underline{a, p}$

$$\hat{x}\hat{p} = \hat{p}\hat{x}$$

2.2. Mathematical Notions: Hilbert spaces & operators.

Def A Hilbert space H is a

① complete complex vector space

$$\psi, \phi \in H, a, b \in \mathbb{C} \Rightarrow a\psi + b\phi \in H.$$

② Inner product in H .

$$\langle \cdot | \cdot \rangle : H \times H \rightarrow \mathbb{C}.$$

Conjugate linear $(\psi, \phi) \mapsto \langle \psi | \phi \rangle$

$$(i) \langle \psi | \phi \rangle = \overline{\langle \phi | \psi \rangle}, (ii) \langle \psi | \psi \rangle \geq 0, (iii) \text{equality only holds for } \psi = 0.$$

$$(iv) \langle \psi | a\phi_1 + b\phi_2 \rangle = a\langle \psi | \phi_1 \rangle + b\langle \psi | \phi_2 \rangle$$

$$\|\psi\| = \sqrt{\langle \psi | \psi \rangle} \rightarrow \text{Induced norm.}$$

④ $\gamma \in H$ is a unit vector if $\|\gamma\| = 1$.

⑤ $\langle \gamma | \varphi \rangle = 0 \Rightarrow \gamma, \varphi$ are orthogonal.

⑥ $H = \text{span}\{e_j | j \in I\} \Rightarrow \{e_j\} \rightarrow \text{Basis of } H$.

⑦ If $\{e_j | j \in I\}$ is a basis and $\langle e_j | e_k \rangle = \delta_{jk}$
 $\Rightarrow \{e_j\}$ is an orthonormal basis (ONB).

⑧ If H has a countable basis $\Rightarrow$ separable.

Examples ① Usual scalar product.

$$z, w \in \mathbb{C}^d = H$$

$$\Rightarrow \langle z | w \rangle = \sum_{j=1}^d \overline{z_j} w_j$$

$\Uparrow$

Physics, this book

Mathematical text book
 $\downarrow$
 $\langle z | w \rangle = \sum_{j=1}^d z_j \overline{w_j}$

② Lebesgue measure.

$$\gamma_1, \gamma_2 \in L^2(\mathbb{R}^3)$$

From Cauchy-S Inequality

$$\Rightarrow \langle \gamma_1 | \gamma_2 \rangle = \int_{\mathbb{R}^3} \overline{\gamma_1(x)} \gamma_2(x) dx < \infty$$

Remark If $\gamma \in H$ and ONB $\{e_j\}$ are given

$$\Rightarrow \gamma = \sum_j \langle e_j | \gamma \rangle e_j$$

$$\begin{aligned} \langle \gamma | \gamma \rangle &= \sum_{j,k} \langle \underbrace{e_j | \gamma} \rangle e_j, \underbrace{e_k | \gamma} \rangle e_k \rangle = \sum_{j,k} \langle e_j | \gamma \rangle \langle \gamma | e_j \rangle \overbrace{\langle e_i | e_k \rangle}^{\delta_{jk}} \\ &= \sum_j \langle e_j | \gamma \rangle \langle \gamma | e_j \rangle = \sum_j |\langle \gamma | e_j \rangle|^2 \end{aligned}$$

Dirac's Bra-Ket notation.

$\langle \cdot | \cdot \rangle = \text{Bracket}$.
 $\downarrow$ $\downarrow$
 Bra Ket

$$\boxed{\psi_1, \psi_2 \in H} \Rightarrow \underbrace{\langle \psi_1 |}_{\text{Bra}} \underbrace{|\psi_2 \rangle}_{\text{Ket}} = \underbrace{(\langle \psi_1 |)}_{\text{Bra-vector}} \underbrace{(|\psi_2 \rangle)}_{\text{Ket vector.}}$$

$|\psi\rangle \in H$. (Hilbert).

$\langle \psi | \Rightarrow$ Linear functional defined on H . $\langle \psi | \in H^*$
 $\downarrow$
 dual space of H .
 $\forall |\phi\rangle \in H, \underbrace{\langle \psi | \phi \rangle}_{\text{scalar}} \in \mathbb{C}$

ONB $\{|e_j\rangle\} \subset H \Rightarrow \underbrace{\{\langle e_j | \} : \text{dual basis of } H^*}$.

$$\begin{aligned} |e_j\rangle^* &= \langle e_j | & * : H &\rightarrow H^* : \text{Linear map} \\ |e_j\rangle &\rightarrow \langle e_j | \\ \mathbb{C}|e_j\rangle &\rightarrow \mathbb{C}^* \langle e_j | \end{aligned}$$

$$|\psi\rangle \in H \Rightarrow |\psi\rangle = \sum_j a_j |e_j\rangle \iff \langle \psi | = \sum_j \overline{a_j} \langle e_j |$$

$$|\phi\rangle = \sum_j b_j |e_j\rangle \iff \langle \phi | = \sum_j \overline{b_j} \langle e_j |$$

$$\begin{aligned} \langle \phi | \psi \rangle &= \sum_{j,k} (\overline{b_j} \langle e_j |) (a_k |e_k\rangle) \\ &= \sum_{j,k} \overline{b_j} a_k \underbrace{\langle e_j | e_k \rangle}_{\delta_{jk}} = \sum_j \overline{b_j} a_j = \underbrace{\langle \overline{b}, a \rangle}_{\text{Vector (Infinite dim)}} \end{aligned}$$

$$|e_j\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \rightarrow \text{1}^{\text{th}} \text{ component (column vector)}$$

$$|e_j\rangle^* = \langle e_j| = (0, 0, \dots, 1, \dots, 0) \text{ (row vector)}$$

$$\Rightarrow \langle e_j | e_j \rangle = \underbrace{(\dots \quad 1 \quad \dots)}_{1 \times n \text{ matrix}} \underbrace{\begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}}_{n \times 1 \text{ matrix}} = 1$$

$\star \Rightarrow$ adjoint operation

A : operator on $\mathcal{H}$.

$A: \mathcal{H} \rightarrow \mathcal{H}$ $\wedge$ ONB $\{e_j\}$

$$|\psi\rangle \in \mathcal{H}, \quad |\psi\rangle = \sum_j a_j |e_j\rangle$$

$$|\psi\rangle = \sum_j \underbrace{a_j}_{\langle e_j | \psi \rangle} |e_j\rangle = \sum_j |e_j\rangle \underbrace{\langle e_j | \psi \rangle}_{\substack{\text{1} \times \text{n}, \text{n} \times \text{n}, \text{n} \times \text{1}}} \\ \langle e_j | \psi \rangle = \left(\sum_j |e_j\rangle \langle e_j| \right) |\psi\rangle \quad \forall |\psi\rangle \in \mathcal{H}$$

$\star$ Identity map $|\psi\rangle \rightarrow |\psi\rangle$

$\star$ $\boxed{\mathbb{I}_{\mathcal{H}} = \sum_j |e_j\rangle \langle e_j|} \rightarrow$ Important Identity In Q.M.

e.g. $\sum_j |e_j\rangle \langle e_j| = |e_1\rangle \langle e_1| + |e_2\rangle \langle e_2| + |e_3\rangle \langle e_3|$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3$$

$$A = \mathbb{I}_H A \mathbb{I}_H = \sum_{j,k} |e_j\rangle \langle e_j| A |e_k\rangle \langle e_k|$$

$$(\langle e_j | A | e_k \rangle \Rightarrow \text{scalar}, A_{jk} = \underbrace{\langle e_j |}_{H^*} \underbrace{A | e_k \rangle}_H)$$

$$= \sum_{j,k} A_{jk} |e_j\rangle \langle e_k| \Rightarrow \text{Matrix representation of } A.$$

Examples ① $|\psi\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, |\phi\rangle = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$

$$\Rightarrow |\psi\rangle \langle \phi| = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} (\bar{b}_1, \dots, \bar{b}_n) = \begin{pmatrix} a_1 \bar{b}_1 & \dots & a_1 \bar{b}_n \\ \vdots & & \vdots \\ a_n \bar{b}_1 & \dots & a_n \bar{b}_n \end{pmatrix}$$

② Let $H \in \mathbb{R}^2$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow \langle 0| = (1 \ 0), \langle 1| = (0 \ 1)$$

$$\Rightarrow \left\{ \begin{aligned} |0\rangle \langle 0| &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, |1\rangle \langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, |0\rangle \langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \\ |1\rangle \langle 1| &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned} \right\} \Rightarrow \text{forms a basis of } \mathbb{C}^{2 \times 2} = H \times H.$$

Def (Adjoint Map).

① A linear map $A: H \rightarrow H$ is called an operator on H .

② The set of all operators on $H = L(H)$.

③ $T: L(H) \rightarrow L(H) \Rightarrow T$ is called a super-operator.

④ $A^*: H \rightarrow H$ is called the adjoint operator to A if

$$\langle A^* \psi | \phi \rangle = \langle \psi | A \phi \rangle \quad \forall |\phi\rangle, |\psi\rangle \in H.$$

⑤ If $A = A^* \Rightarrow$ self-adjoint (Hermitian).

Matrix representation of A^*

Conclusion is $[A^*]_{ij} = [A]_{ji}^*$ →

$$|\psi\rangle = \sum_i a_i |e_i\rangle, \quad |\varphi\rangle = \sum_j b_j |e_j\rangle$$

$$\langle \psi | A | \varphi \rangle = \langle A^* \psi | \varphi \rangle$$

$$\begin{aligned} \langle e_i | A | e_j \rangle &= A_{ij} \\ \langle e_i | A^* | e_j \rangle &= A_{ji}^* \end{aligned}$$

$$A = \sum_{i,j} |e_i\rangle A_{ij} \langle e_j|$$

L.H.S

$$\langle \psi | A | \varphi \rangle = (\langle \psi |) \cdot A \cdot (|\varphi\rangle) = \left(\sum_i \bar{a}_i \langle e_i | \right) \left(\sum_{j,k} |e_j\rangle A_{jk} \langle e_k| \right) \left(\sum_l b_l |e_l\rangle \right)$$

$$= \sum_{i,j,k,l} \bar{a}_i A_{jk} b_l \underbrace{\langle e_i | e_j \rangle}_{\delta_{ij}} \underbrace{\langle e_k | e_l \rangle}_{\delta_{kl}}$$

$$= \sum_{i,l} \bar{a}_i A_{li} b_l$$

$$\langle A^* \psi | \varphi \rangle = \sum_{j,k} \bar{A}_{jk}^* \bar{a}_k b_j = \sum_{i,l} \bar{A}_{li}^* \bar{a}_i b_l$$

$$\Rightarrow \boxed{A_{li} = \bar{A}_{il}^*}$$

adjoint in Matrix.

Matrix representation of adjoint ... ↗

Def. $U \in L(H)$ is unitary if

$$\langle U\psi | U\varphi \rangle = \langle \psi | \varphi \rangle \quad \forall |\psi\rangle, |\varphi\rangle \in H.$$

$$\left(\begin{aligned} \text{Rank. } \langle U\psi | U\varphi \rangle &= \langle U^* U \psi | \varphi \rangle = \langle \psi | \varphi \rangle \\ &\Rightarrow \boxed{U^* U = \mathbb{I}_H} \end{aligned} \right)$$

② The set of unitary operators on $H = \mathcal{U}(H)$.

Def $A \in L(H)$, $|\psi\rangle \in H \setminus \{0\}$.

$$\textcircled{1} A|\psi\rangle = \lambda |\psi\rangle \Rightarrow \begin{cases} \lambda : \text{eigen value} \\ |\psi\rangle : \text{eigen vector. (state)} \end{cases}$$

$$\textcircled{2} E_\lambda = \text{span} \{ |\psi_1\rangle, \dots, |\psi_n\rangle \}$$

" eigen vectors w.r.t. λ .

eigen span.

$$\textcircled{3} \lambda \text{ is non-degenerate if } \dim E_\lambda = 1$$

$$\textcircled{4} \lambda \text{ is degenerate if } \dim E_\lambda \neq 1. (>1)$$

$$\textcircled{5} \sigma(A) = \{ \lambda \in \mathbb{C} : (A - \lambda I)^{-1} \text{ does not exist} \}$$

$\hookrightarrow$ Spectrum of the operator A .

* Self-Adjoint operators are diagonalizable.

$\Rightarrow \exists$ orthonormal basis $\{|e_{j,\alpha}\rangle\}$ s.t.

$$A|e_{j,\alpha}\rangle = \lambda_j |e_{j,\alpha}\rangle$$

We put α , since eigen vectors can not be unique.

A : self-adjoint operator with $|e_{j,\alpha}\rangle$ with $\langle e_{j,\alpha} | e_{k,\beta} \rangle = \delta_{j,k} \delta_{\alpha,\beta}$.

$$\Rightarrow A_{j,\alpha; k,\beta} = \langle e_{j,\alpha} | A | e_{k,\beta} \rangle = \langle e_{j,\alpha} | \lambda_k | e_{k,\beta} \rangle \quad \text{if } \alpha=k, \alpha=\beta$$

$$A = \sum_{j,\alpha,k,\beta} |e_{j,\alpha}\rangle \langle e_{j,\alpha} | A | e_{k,\beta} \rangle \langle e_{k,\beta} | = \lambda_k \langle e_{j,\alpha} | e_{k,\beta} \rangle = \lambda_k \delta_{j,k} \delta_{\alpha,\beta}$$

$$= \sum_{j,\alpha,k,\beta} \lambda_k \delta_{j,k} \delta_{\alpha,\beta} |e_{j,\alpha}\rangle \langle e_{k,\beta}| = \sum_{j,\alpha} \lambda_j |e_{j,\alpha}\rangle \langle e_{j,\alpha}|$$

diagonal form $\rightarrow$

$$\text{From } A = \sum_{j,\alpha} \lambda_j |e_{j,\alpha}\rangle \langle e_{j,\alpha}|, \quad |\gamma\rangle = \sum_{k,\beta} a_{k,\beta} |e_{k,\beta}\rangle$$

$$\Rightarrow \langle \gamma | A | \gamma \rangle = \sum_{\substack{j,\alpha \\ k,\beta \\ l,\gamma}} \overline{a_{j,\alpha}} \langle e_{j,\alpha} | \left(\lambda_k |e_{k,\beta}\rangle \langle e_{k,\beta}| \right) (a_{l,\gamma} |e_{l,\gamma}\rangle)$$

$$= \sum_{\substack{j,\alpha,k,\beta \\ l,\gamma}} \overline{a_{j,\alpha}} \lambda_k a_{l,\gamma} \delta_{jk} \delta_{\alpha\beta} \delta_{kl} \delta_{\gamma\alpha}$$

$$= \sum_{j,\alpha} \overline{a_{j,\alpha}} \lambda_j a_{j,\alpha} = \sum_{j,\alpha} \lambda_j |a_{j,\alpha}|^2$$

$$\Rightarrow \boxed{\lambda_{\min} \|\gamma\|^2 \leq \langle \gamma | A | \gamma \rangle \leq \sum_{j,\alpha} \lambda_{\max} |a_{j,\alpha}|^2 = \lambda_{\max} \|\gamma\|^2}$$

Def (1) $P \in L(H)$ is a projection iff $P^2 = P$.

(2) $P_{\text{sub}} |\psi\rangle = |\psi\rangle \quad \forall |\psi\rangle \in H_{\text{sub}}$: subspace of H .

$\Rightarrow P_{\text{sub}}$ is a projection onto this subspace.

Example (1) $P_j = \sum_{\alpha=1}^{d_j} |e_{j,\alpha}\rangle \langle e_{j,\alpha}|$: Projection onto E_{λ_j} .

$$\textcircled{2} \boxed{P_j = |e_j\rangle \langle e_j|}$$

$$\Rightarrow P_j P_k = |e_j\rangle \langle e_j | e_k \rangle \langle e_k| = |e_j\rangle \langle e_k| \delta_{jk} \\ = |e_j\rangle \langle e_j| \delta_{jk} = P_j \delta_{jk}.$$

(3) $A = \sum_j \lambda_j \textcircled{P_j} \rightarrow e_j$ is an eigenstate w.r.t. λ_j .

Def ① $A \in L(H)$

$$\Rightarrow \|A\|_{op} (= \|A\|) = \sup \{ \|A\psi\| : \|\psi\|=1 \}.$$

↑
this book.

iff $\|A\|_{op} < \infty \Rightarrow A$ is bounded.

② A is positive iff $\langle \psi | A \psi \rangle \geq 0 \quad \forall |\psi\rangle \in H$

③ A is strictly positive iff $\langle \psi | A \psi \rangle > 0 \quad \forall |\psi\rangle \in H \setminus \{0\}$.

④ A, B are commute iff $[A, B] = AB - BA = 0$.

Recall $\lambda_{\min} \|\psi\|^2 \leq \langle \psi | A | \psi \rangle \leq \lambda_{\max} \|\psi\|^2$

$\Rightarrow$ ① $\|A\|_{op} = \lambda_{\max}$

② A is positive iff $\lambda_{\min} \geq 0$

③ A is strictly positive iff $\lambda_{\min} > 0$.

Def $\{|e_j\rangle\} : \text{ONB in } H$

The trace is

$$\text{tr} : L(H) \rightarrow \mathbb{C}$$

$$A \mapsto \text{tr}(A) = \sum_j \langle e_j | A e_j \rangle = \sum_j A_{jj}$$

Since it does not depend on choice of ONB $\{|e_j\rangle\}$, trace was well-defined.

↓

2.2

Reference: (Griffith) Introduction to Quantum Mechanics.

$$1D \text{ Schrödinger Equation} \Rightarrow \hat{H}|\psi\rangle = \hat{H}|\psi\rangle$$

↗ Hamiltonian operator.

In classical mechanics, $H(x, p) = \underbrace{\frac{p^2}{2m}}_{\text{Kinetic Energy}} + \underbrace{V(x)}_{\text{potential Energy}}.$

$$\hat{H}(\hat{x}, \hat{p}) = \frac{\hat{p}^2}{2m} + V(\hat{x}). \quad (\hat{p}, \hat{x} \text{ to express Hamiltonian operator})$$

↳ Hermitian

$$\hat{x}|\alpha\rangle = \alpha|\alpha\rangle \Rightarrow \begin{cases} \alpha: \text{eigen value of position operator} \\ |\alpha\rangle: \text{eigen state of } \hat{x}. \end{cases}$$

$$\hat{p}|\alpha\rangle = \beta|\alpha\rangle \Rightarrow \begin{cases} \beta & \text{---} \\ |\alpha\rangle & \text{---} \end{cases} \quad (\hat{p})$$

In Linear Algebra Class we studied

A, B are simultaneously diagonalizable $\Leftrightarrow [A, B] = 0.$

(Q) Does $\hat{x}, \hat{p}$ commute? (A) No!

If it's true, $\exists |\psi\rangle$ s.t.

$$\begin{cases} \hat{x}|\psi\rangle = \alpha|\psi\rangle \\ \hat{p}|\psi\rangle = \beta|\psi\rangle \end{cases} \rightarrow \text{We can obtain exact value of position and momentum simultaneously.}$$

$\Rightarrow$ Heisenberg's Uncertainty Principle $\langle \hat{x} \rangle \langle \hat{p} \rangle \geq \hbar/2.$

We can express $\hat{p} = \frac{\hbar}{i} \partial_x$. (in position basis)

$$\langle x | \hat{H} | \psi \rangle = \langle x | \frac{\hat{p}^2}{2m} + V(x) | \psi \rangle$$

eigen bra-vector of $\hat{x}$

$$= \frac{1}{2m} \langle x | \hat{p}^2 | \psi \rangle + \langle x | V(x) | \psi \rangle$$

$\hat{p}^2 \xrightarrow{-\hbar^2 \partial_x^2}$

$(V(x) | x \rangle) = V(x) | x \rangle$ eigen vector with eigenvalue x of $\hat{x}$.

(infinite polynomial of $\hat{x}$)

$$= -\frac{\hbar^2}{2m} \partial_x^2 \langle x | \psi \rangle + V(x) \langle x | \psi \rangle$$

If we set $\psi(x,t) = \langle x | \psi \rangle$,

$$i\hbar \partial_t |\psi\rangle = \hat{H} |\psi\rangle \Rightarrow i\hbar \partial_t \langle x | \psi \rangle = \langle x | \hat{H} | \psi \rangle$$

$$\Rightarrow i\hbar \partial_t \psi(x,t) = -\frac{\hbar^2}{2m} \partial_x^2 \psi(x,t) + V(x) \psi(x,t)$$

Well-known form of Schrödinger Eq.

Meaning of $\psi(x,t) = \langle x | \psi \rangle$

$|x\rangle$: orthonormal basis

State contains lots of information
position, momentum, ...

$$|\psi\rangle = \int |x\rangle \langle x | \psi \rangle dx$$

discrete version $\Rightarrow \mathbb{I}_H = \sum_j |e_j\rangle \langle e_j|$
continuous version $\Rightarrow \mathbb{I}_H = \int |x\rangle \langle x| dx$

$$\langle \psi | \psi \rangle = \int \langle \psi | x \rangle \langle x | \psi \rangle dx = \int |\langle \psi | x \rangle|^2 dx$$

$|\psi(x,t)|^2$

$\langle \psi | x \rangle \langle x | \psi \rangle \xrightarrow{\delta(x-y)}$

Probability density function over $x \in \mathbb{R}^1$.

$$[\hat{x}, \hat{p}] = ??$$

$$\begin{aligned} [\hat{x}, \hat{p}] f(x) &= \hat{x} \hat{p} f(x) - \hat{p} \hat{x} f(x) \\ &= \hat{x} \left(+\frac{\hbar}{i} \partial_x f(x) \right) - \left(+\frac{\hbar}{i} \partial_x \right) (\hat{x} f(x)) \\ &= +\frac{\hbar}{i} x \partial_x f(x) - \frac{\hbar}{i} \partial_x (x f(x)) \\ &= -\frac{\hbar}{i} f(x), = i\hbar f(x). \end{aligned}$$

$$\Rightarrow [\hat{x}, \hat{p}] = i\hbar$$

(eg) $[\hat{x}, \hat{H}]$

$$\begin{aligned} &= [\hat{x}, \frac{\hat{p}^2}{2m} + V(\hat{x})] = \frac{1}{2m} [\hat{x}, \hat{p}^2] \\ &\quad \text{commutes} \\ &= \frac{1}{2m} (\hat{x} \hat{p}^2 - \hat{p}^2 \hat{x}) \\ &= \frac{1}{2m} ([\hat{x}, \hat{p}] \hat{p} + \hat{p} [\hat{x}, \hat{p}]) = \frac{i\hbar}{m} \hat{p} \end{aligned}$$

$$\begin{aligned} &[\hat{x}, \hat{p}] = i\hbar \neq 0 \\ &\hat{x}, \hat{p} \\ &\underline{V(x)=0} \\ &[\hat{p}, \hat{H}] = 0 \rightarrow \end{aligned}$$

